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Normalization of Abbreviation and Acronym on Microtext in Bahasa Indonesia by Using Dictionary-Based and Longest Common Subsequence (LCS)

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Abstract

The communication nowadays has reached a need to express the idea in short text. This kind of communication is delivered in various media such as short messages service (SMS), Facebook status, Twitter post, chat messages, comments, and any form of short text. These various kinds of short text are known as microtext. The microtext usually has one sentence or less, written informally, consists of abbreviations, acronyms, emoticons, hashtags, and others. These features of the microtext become a particular challenge to the text classification. These features cannot be processed directly as in the traditional text processing, because it may lead to inaccuracy. Therefore, it requires microtext normalization to transform these features into well-written texts before applying text processing. This research aims to normalize some of these features, which are abbreviations and acronyms. The normalization applied dictionary-based and longest common subsequence (LCS) approach to the microtext in Bahasa Indonesia. Dictionary-based has shown an excellent performance instead of LCS. However, it is limited to pre-defined abbreviations and acronyms. Besides, the LCS offers dynamic microtext normalization. Nevertheless, the combination of both approaches increases normalization performance slightly.

Keywords: microtext normalization; abbreviation; acronym; dictionary-based normalization; longest common subsequence normalization;

1. Introduction

The term "microtext" refers to the brief and semi-structured text that has an informal manner and written in one sentence or less [1]. Most of the microtext, such as short messaging service (SMS), comments, chat or microblog contains abbreviation, emoticons, hashtags, typographical errors, and others. As a result, processing micro-text requires more effort to obtain the information inside it [2]. Generally, the research that used Twitter or SMS as the input applies microtext normalization process before conducting the text processing. It transforms the non-standard

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texts into well-written texts. Therefore, the traditional approach on text processing can be made [3]. Previous research conducted manual microtext normalization for Twitter post and SMS spam filtering to obtain a good result [4][5]. The microtext normalization will contribute to yield the same keyword among texts that used in the bag-of-words (BOW) method for text classification or clustering. Some previously conducted research relied on bag-of-words method to classify or cluster the text [6][7]. Manual normalization is not adequate for massive data and should be avoided. Therefore, the automatic microtext normalization is required to normalize a massive text data. The normalization improves classification accuracy in the field of sentiment analysis [8].

In 2011, research by Xue [9] proposed a normalization approach which incorporates four factors, such as phonetic, orthographic, acronym expansion, and contextual factor. They conducted the normalization by using the channel model approach for each element. They found that the method was able to outperform existing algorithms on a public SMS data set instead of twitter posts. This research considered the acronym channel as the complementary model. It uses a dictionary to perform acronym normalization. They collected the set of popular acronyms from a public website to build the dictionary.

Furthermore, another research uses a different approach. The microtext normalization is done using probably-phonetically-similar word discovery [10]. This proposed algorithm required training to determine pronunciation possibility of English words based on spelling. The new out of vocabulary (OOV) will identify the most probable in-vocabulary (IV) words with similar pronunciation. The accuracy of this research for certain types of normalization lies in the range of 80%-90%. However, the research performs with the accuracy of almost 60%. Meanwhile, another research adapted Spanish microtext normalization to English [11]. They suggested POS tagging as a new process to improve the candidate selection.

In 2017, research was conducted to normalize twitter posts, which are published in Bahasa Indonesia [2]. This research conducted in three stages, such as cleaning process, OOV detection, and word replacement. This research requires a dictionary to detect OOV. They built the dictionary by collecting the slang words from their dataset. This research also uses context-dictionary to solve the ambiguity problem.

Previous research [9][2] built their dictionary to handle acronym. Our research uses a similar approach. However, instead of creating a new dictionary from a limited dataset, we make use of the existing dictionary that provides common abbreviation and acronym. Besides, we also utilize the longest common subsequence (LCS) method to handle the abbreviation and acronym that does not exist in the dictionary.

The microtext contains unstructured text which can have many grammatical looses such as uncommon abbreviation. Also, microtext includes acronym to shorten the number of typed characters. Therefore, this research focuses on abbreviation and acronym normalization in microtext. We proposed the dictionary-based combined with longest common subsequence (LCS) method.

This paper is organized as follow: the first section is an introduction that discusses the background and the aim of the research. The second section describes the proposed method to normalize the abbreviation and acronym. The third section discusses the result of the experiment and the whole research is summarized in the last section.

2. Research method

The research is divided into three phases. The first phase is the data collection. This research uses microtext as the data source. The second phase is the text pre-processing, which prepares the text before being normalized. The third phase normalized each word in the twitter posts. Fig 1 illustrates all phases of this research.

2.1. Data collection

The first phase is collecting the microtext. This research use twitter post as the microtext. The data source is collected by crawling them. This research removes the twitter posts, which consist of several words only. The twitter posts are also selected manually to obtain the twitter posts which contain abbreviation or acronym. The number of collected twitter posts are 400.

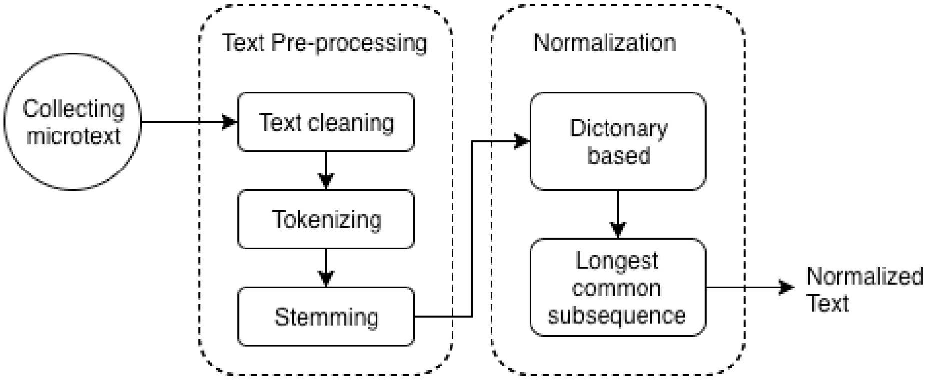


Fig. 1. General architecture of normalization process.

2.2. *Text pre-processing*

This research proposes the normalization method a twitter post by using the Longest Common Subsequence (LCS) and dictionary-based. Fig. 1 illustrates the general architecture of the normalization process. It begins with the text pre-processing. The first task of text pre-processing is text cleaning. This task is done to ensure the unused items such as punctuations, retweet (the act where someone re-post other twitter posts), links, or hashtag (one or more keywords or phrases to identify post in a social media, which precede by the # sign). For example, Fig. 2a shows the original twitter post. Meanwhile, the 2b shows the clean version.

RT awas tertipu dg racun para salesman amatir! mrk sanggup blg apa aja spy target terpenuhi! hati2 gaes #Salesman #Penipu

(a) Original twitter post

awas tertipu dg racun para salesman amatir mrk sanggup blg apa aja spy target terpenuhi hati2 gaes

(b) Cleaned twitter post

Fig. 2. The text pre-processing on a twitter post.

The second task of text pre-processing is tokenizing. This task will deduct the text into the smallest parts called token. For example, the text in Fig. 2b will be tokenized to ”awas”, ”tertipu”, ”dg”, ”racun”, ”para”, ”salesman”, ”amatir”, ”mrk”, ”sanggup”, ”blg”, ”apa”, ”aja”, ”spy”, ”target”, ”terpenuhi”, ”hati2”, and ”gaes”.

The last text pre-processing task is stemming. Stemming is the process to extract a token into its root word. For example, the token ”tertipu” will be ”tipu” after the stemming task. The word ”ter” is a prefix of the token ”tertipu”. This task is required to find the correct word. The correct word means that the stemming process successfully identifies that the token has a root word or the token itself is a root word. Therefore, the stemming process requires a dictionary to ensure the validity of a token. In other words, the research applies the normalization process to the tokens that are not recognized in the dictionary. The correct words (the root word or the word with affix) are left as is and will not be normalized.

The stemming process may reduce misidentification. For example, the token ”satu” is a root word that has a particular meaning (in English: one). If the LCS algorithm processes this token, it will find the matched words such as ”satu”, ”saturnus”, ”saturasi”, ”pemersatu”, ”shiatsu”, ”sepatu”, ”salah satu” and every word or phrase that consists of ”s”, ”a”, ”t”, and ”u” in sequence. All of these matched words are not the normalized version of the token satu (except the word ”satu”). The process of matching token ”satu” to the words in the dictionary is wasting CPU and memory resources, because ”satu” is the correct term. This issue also occurs to the word with the affix. Therefore, by implementing the stemming task in the text pre-processing, the tokens are no longer required to be matched with the words in the dictionary, as well as applying the LCS algorithm. As a result, the usage of CPU and memory resources can be reduced. Moreover, we estimate that the required time to complete the normalization process will be faster.

Table 1. The excerpt of dictionary-based abbreviation and acronym.

No.	Abbreviation		Acronym	
	Entries	Full form	Entries	Full form
1	yg	yang	asi	air susu ibu
2	tdk	tidak	sidak	inspeksi mendadak
3	pd	pada	hut	hari ulang tahun
4	tp	tapi	sim	surat izin mengemudi
5	kls	kelas	wagub	wakil gubernur
6	hlm	halaman	perwal	peraturan walikota
7	ktp	kartu tanda penduduk	walubi	wali umat buddha indonesia
8	dpr	dewan perwakilan rakyat	kapolri	kepala kepolisian republik indonesia
9	tki	tenaga kerja indonesia	pemilu	pemilihan umum
10	yth	yang terhormat	tilang	bukti pelanggaran

Table 2. The excerpt of unlisted abbreviation.

No.	Entries	Correct Word
1	ftkp	fotokopi
2	nmpk	nampak
3	mbl	mobil
4	br	baru
5	mnsia	manusia
6	td	tadi
7	klmpk	kelompok
8	slmt	selamat
9	plg	pulang
10	kntr	kantor

2.3. Normalization

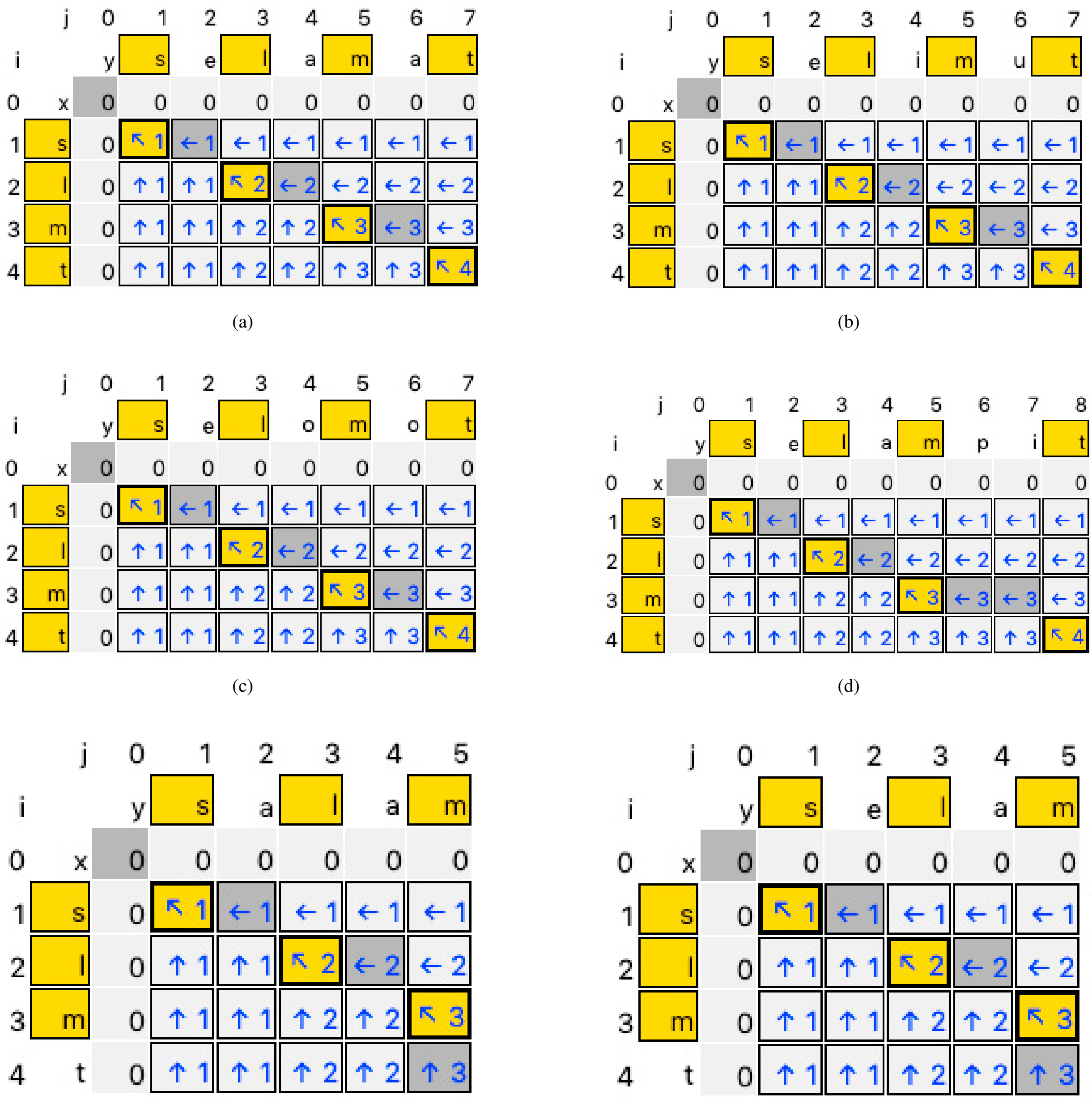
The primary process of this research is normalization. The normalization process utilizes a dictionary that provides common abbreviation and acronym. The dictionary is provided by Kateglo. It contains more than 7,000 abbreviations and acronyms. Those include the official abbreviation and acronym by the Indonesian dictionary and also the common abbreviation used in daily life. Table 1 shows the example of the abbreviation and acronym entries.

Although the dictionary provides many abbreviations and acronyms, the twitter posts might consist of the unpredictable and unlisted abbreviation. For example, the abbreviations in Table 2 are not available in the dictionary. Nevertheless, most of the Internet users used them regularly.

The unpredictable and uncommon abbreviation is the issue that raised for the dictionary-based approach for normalization. Therefore, this research also applies the longest common subsequence (LCS) to provide the microtext normalization that aside from the dictionary. LCS finds the common subsequence in a string. On the contrary of the longest common substring, the subsequence is not required to occupy the original position consecutively.

For example, consider there is a token ”slmt” from the twitter post. After the stemming process, this token is not recognized as the existing word in the dictionary. Therefore, this token will be taken to the next process, which calculates the common subsequence between two strings. The first string is the token, and the second string is obtained from the dictionary. All of the words in the dictionary will be examined one by one with the first string. However, because the first letter of the abbreviation will be the first letter of the word, then the examination only occur to the words that have the same first letter with the abbreviation. As an example, the token ”slmt” begins with the letter ”s”. Hence the second string should start with the letter ”s” as well. Other words that do not start with the letter ”s” will be ignored. This research uses the dictionary provided by Kateglo.

The output of the common subsequence searching might vary, depends on the available word in the dictionary. For example, the token slmt matches with several words, such as selamat, selimut, selampit, and selimpat. Fig. 3 illustrates the process to determine the common subsequence. Fig. 3a shows the function backtracking. It would follow from the bottom right to the left corner when searching for the common subsequence. The yellow background cell is part of the LCS.



(a) (b) (c) (d) (e) (f)

Fig. 3. Common subsequence searching process.

There is a possibility that there is the number of words that contain ”slmt” in sequence, such as ”selamat”, ”selimut”, ”solomot”, and ”selampit” as shown in Fig. 3(a-d). In addition, the LCS also recognize that the words ”selam” and ”salam” (shown in Fig. 3(e-f)) have the common subsequence ”slm”. However, this research determines that microtext normalization process should strictly use the ”slmt” as the acceptable common subsequence. As a result, other common subsequences will be ignored.

Table 3. Testing Result.

Method	TP	TN	FN	FP	Accuracy
Dictionary-based	3957	587	941	53	82%
LCS	3579	566	1048	375	74%
Dictionary-based + LCS	3682	984	536	366	84%

3. Result and discussion

The microtext normalization by using dictionary-based and longest common subsequence (LCS) was conducted on 400 twitter posts. This research was tested on the twitter post in Bahasa Indonesia. We conducted three experiments in the microtext normalization. First, we examined the normalization process by using the dictionary only. Second, we substitute the microtext normalization by using the LCS algorithm. The last experiment combines both methods by applying microtext normalization using the dictionary method, then utilizing the LCS algorithm if there is no matching word in the dictionary. Table 3 shows the result of all the experiments. We calculate the accuracy by using the formula 1.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

This research determines the true positive or TP as the word that should be left as is and does not require normalization process. There are two kinds of words, such as the root word and the word with the suffix. The root word of both types should exist in the dictionary. For example, the word ”satu” does not have affix because it is a root word. As the word ”satu” is a root word, it should exist in the dictionary.

Another example is the word ”berlari”. The word ”berlari” does not exist in the dictionary, because the word ”berlari” is formed by ”ber” (prefix) + ”lari” (root word). Therefore, the word ”lari” should exist in the dictionary. The evaluation result should be true if both words ”satu” and ”berlari” are not changed.

The true negative or TN determines the abbreviation or acronym that should be normalized to the correct word. For example, ”yg” and ”ftkp” should be normalized to ”yang” and ”fotokopi” respectively. There is a possibility that more than one word is suggested as the output. In this case, we consider this issue as incorrect normalization.

The false negative or FN determines the abbreviation or acronym that falsely normalized. For example, the term ”hlm” usually refers to ”halaman”. However, the longest common subsequence obtains ”halim” as a result. The last is false positive, or FP. False positive means that the term is already correct, but the system considers it as an incorrect result. For example, by using the LCS algorithm, the term ”banyakin” is matched with ”banyak sedikitnya” because the dictionary contains the phrase ”banyak sedikitnya”. The term ”banyakin” is a slang word. This term is commonly used by the community and cannot be handled by the LCS algorithm.

The first experiment utilized the dictionary. As mentioned in section 2, this research makes use of the Kateglo dictionary, which provides common abbreviation and acronym. This method is the easiest way to perform microtext normalization. We implemented the string matching between the token in a Twitter post to the available abbreviation and acronym in the dictionary. However, we found there are several errors occur because there is more than one meaning in a dictionary entry. For example, Bahasa Indonesia uses the term ”di” is a prefix or a preposition for a place. On the other hand, ”di” is an abbreviation which means ”daerah istimewa” (special region).

Another example is the term ”dr” that is commonly used to determine ”dokter” or ”dari”. However, according to the dictionary, the output is ”dana reboisasi”. This result is incorrect and reduces accuracy. Although the dictionary based has a good result, we estimate the accuracy of this method might increase if the dictionary source is well-prepared. Future research might omit the unrelated abbreviation from the dictionary.

The second experiment implemented the LCS algorithm. This method shows a mediocre result. According to the result, this approach only yields 74% accuracy. Table 3 shows there is 1050 token that is incorrectly identified by the longest common subsequence. We examine that there are many similar words which share the same common subsequence. For example, the subsequence ”bru” might yield ”biru” or ”baru”. This condition yields true or false according to the number of matched letters. Another error occurs when performed the common subsequence of the word ”slmt”. There are more than two words that contain common subsequence ”slmt”, such as ”selamat”, ”selimut”, ”selampit” and so on. As a result, this influence the search result of the common subsequence ”slmt”. Future research should develop a mechanism to predict the most suitable term for the normalization.

The last experiment is the combination between the dictionary-based method with the longest common subsequence (LCS) algorithm. According to Table 3, the result is slightly better with the dictionary-based method. The accuracy increases by two percent after implementing the LCS algorithm. We examine that the dictionary has done most of the work of normalization. In addition, the LCS algorithm does not perform well because of many words has similar subsequence. Therefore, it is required to add another step after the LCS algorithm or substitute the algorithm to obtain better accuracy.

4. Conclusion

Microtext has become a part of communication media. People send microtext in the form of short messaging service (SMS), social media post (Twitter or Facebook), chat message, and anything that has a short body of the text. Because of its limitation in the number of characters, people tend to substitute the word or term into abbreviation or acronym. The abbreviation or acronym may lead to unpredictable text classification result. This research aims to identify the correct term of the abbreviation and acronym. We proposed the dictionary-based and LCS approach to yield the expected result. The data source is obtained from Twitter posts. The normalization by using dictionary-based yield excellent performance. However, the dictionary-based requires high validity data and maintenance cost. The current dictionary contains a large number of entries that are not related to the context. Moreover, the LCS shows a good performance if there is only one possibility for the term. However, there is a lot of similar result which can not be selected automatically. The result shows that the combination of dictionary-based and LCS normalization perform slightly better result than the dictionary-based normalization. This method can be replicated to the microtext in different languages. To replicate this method in different languages, one should adjust the stemming algorithm and the dictionary to the related language. We conclude that the entry of the database plays a significant role to normalize the abbreviation and acronym. Therefore, we suggest rebuilding the current dictionary by filtering the specific entries for future research. Future research that utilizes the LCS algorithm for normalization process should add another step to determine the normalized term if there is more than one subsequence.

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